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Of Mice and MYC Modelling Medulloblastoma

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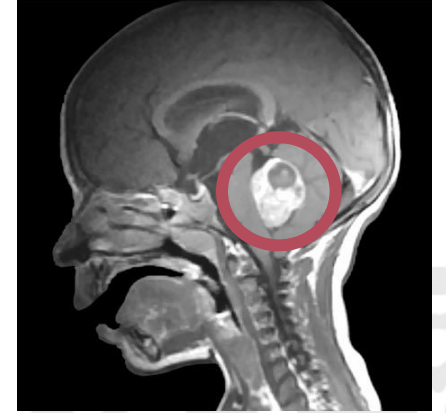
*Name and institution withheld – NDA. Replaced with author's permission





Medulloblastoma

- Most common malignant paediatric brain tumour
~20% of all paediatric brain tumours²
- Embryonal tumour of **hindbrain origin**²
- Approximately **70-80%** long-term survival, depending on risk group, molecular subgroup, age, metastatic status³
- Four **molecular subgroups** – WNT, SHH, Group 3, Group 4



*Sagittal post-contrast
T1-weighted MRI.
Heterogeneously enhancing
fourth ventricular mass*

1. Ostrom QT, et al. Neuro Oncol. 2017. PMID: 29117289
2. Vladoiu MC, et al. Nature. 2019. PMID: 31043743

3. Northcott PA, et al.. Nat. Rev, 2019. PMID: 30765705



Medulloblastoma is a Developmental Cancer

- Medulloblastoma, in most cases, arises due to aberrant developmental programmes¹
- Subgroups are associated with distinct developmental origins/cell states²
- Tumour cells can retain features of normal cerebellar progenitors³
- Developmental signalling pathways can become oncogenic⁴

1. Roussel et al. Curr Top Dev Biol, 2011. PMID: 21295689
2. Hovestadt V, et al. Nat. Rev. Cancer. 2020. PMID: 31819232

3. Cheng et al. Cell Reports, 2020. PMID: 32579914
4. Gilbertson et al. The Lancet Oncology, 2004. PMID: 15050952



Four Main Molecular Subgroups

Wingless (WNT)

- ~10% of cases
- Excellent prognosis
- CTNNb1 pathway alterations

Sonic Hedgehog (SHH)

- ~25-30% of cases
- SHH pathway activation
- Biologically heterogeneous

Group 3

- ~10-20% of cases
- MYC amplification in a subset
- Higher-risk subgroup

Group 4

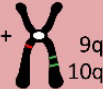
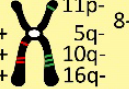
- ~35-40% of cases
- Most common subgroup
- Clinically heterogeneous



Further Subtyping is Possible

Each molecular subgroup can be further subdivided into subtypes

- Optimal subtype classification and stratification remain unresolved

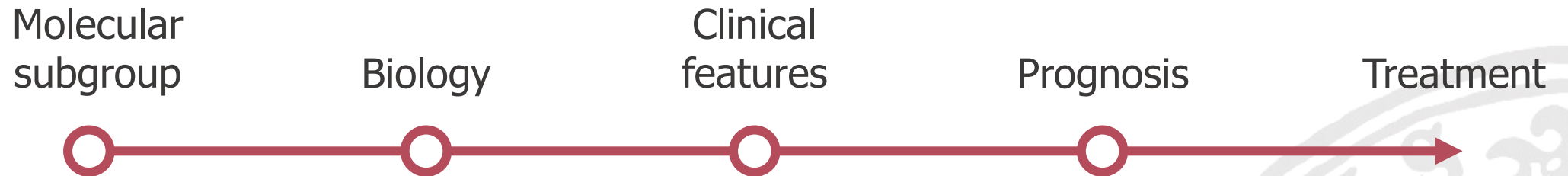
	WNT (9%)		SHH (29%)				Group 3 (19%)			Group 4 (43%)		
	70% 30%		29% 16%		34% 21%		47% 26%		28%		30% 33% 37%	
Subtype	WNTα	WNTβ	SHHα	SHHβ	SHHδ	SHHγ	Gr. 3α	Gr. 3β	Gr. 3γ	Gr. 4α	Gr. 4β	Gr. 4γ
Age	Young	Adult	Young	Infants	Adult	Infants	Infants	Infants/Young		Infants/Young/Adult		
Gender	♂♂ : ♀♀		♂♂ : ♀♀				♂♂ : ♀			♂♂ : ♀		
Prognosis	Very good	Very good	Intermed.	Poor	Intermed.	Good	Better	Poor	Very poor	Intermed.	Intermed.	Intermed.
Histology	Classic, rarely LCA		Desmoplastic/nodular, classic, LCA				Classic, LCA			Classic, LCA		
Metastasis	+		+	++	+	+	++	+	+++	++	++	++
Genomics	Monosomy 6 CTNNb1 mut  WNT signalling		TP53 mut MYCN amp GLI2 amp  PTCH1/SMO/SUFU mut GLI2 amp MYCN amp SHH signalling				i17q 7+ 1q+ 17q+ 18q+ 11p- 5q- 10q- 16q-  Photoreceptor/GABAergic			MYCN amp SNCAIP duplication CDK6 amp i17q 7+ 17q+ 18q+ 11p- X- 8-  Neuronal/Glutamatergic		

Adapted from Taylor et al, 2012 & Cavalli et al, 2017.



Molecular Subgroup $\neq$ Just a Classification

Each of the four main molecular subgroups is biologically and clinically important.



Each subgroup has:

- Distinct cells of origin and developmental programmes
- Different genetic/epigenetic alterations
- Different patterns of dissemination
- Different responses to therapy
- Different clinical outcomes



Despite Improved Survival, Significant Unmet Clinical Needs Remain

Current multimodal treatment

- Surgical resection
- Risk-adapted craniospinal irradiation
- Adjuvant chemotherapy

But...

- Relapse remains largely fatal
- Treatment causes substantial long-term toxicity
- Neurocognitive, endocrine, and other late effects and long-term sequelae are common and of particular importance in children
- Treatment is high risk and metastatic disease remain difficult to treat

There is an unmet need for a better understanding of the disease and for biologically informed, less toxic therapies



Rationale for this Study

Group 3 Medulloblastoma

- Comprises ~20% of all medulloblastoma cases
- Predominantly affects young children
- Poorest prognosis subgroup - <50% 5-year survival
- Characterised by frequent MYC dysregulation/amplification

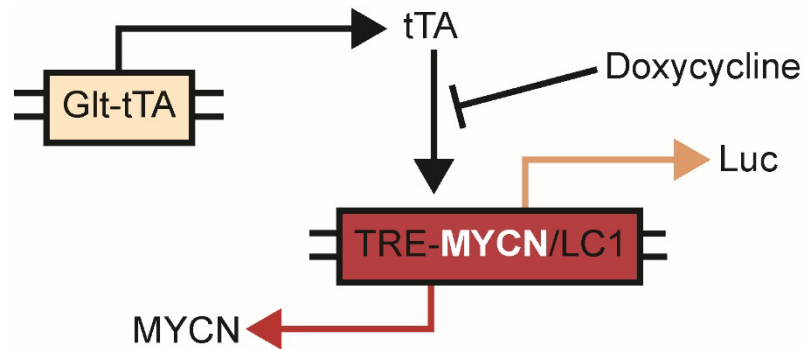
But we know little about its developmental origins and cell of origin, and treatment options remain inadequate.

In this study, we generated and validated a novel, pre-clinical murine model of Group 3 medulloblastoma to investigate its developmental origins, identify the cell(s) of origin, and evaluate potential chemo- and radio-therapy options.

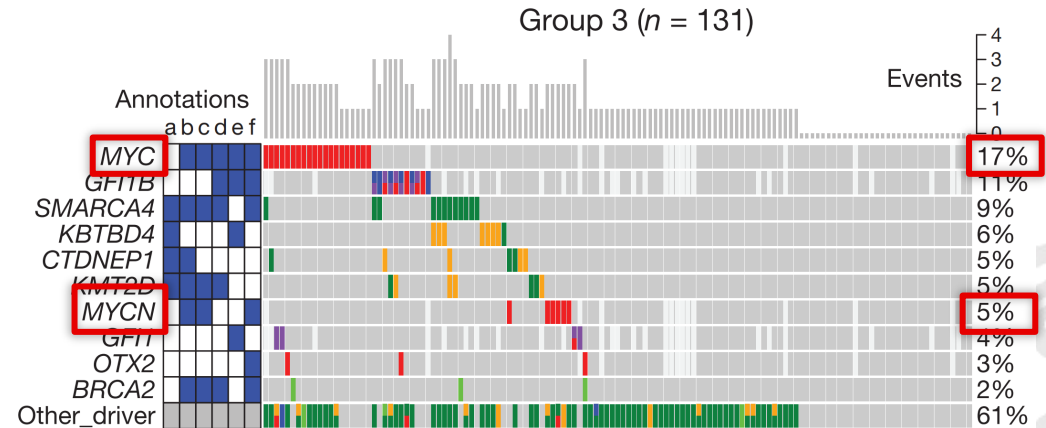


Overview – MYC/MYCN-driven Mouse Models

Original **MYCN**-driven Group 3 Model (**GTML**)

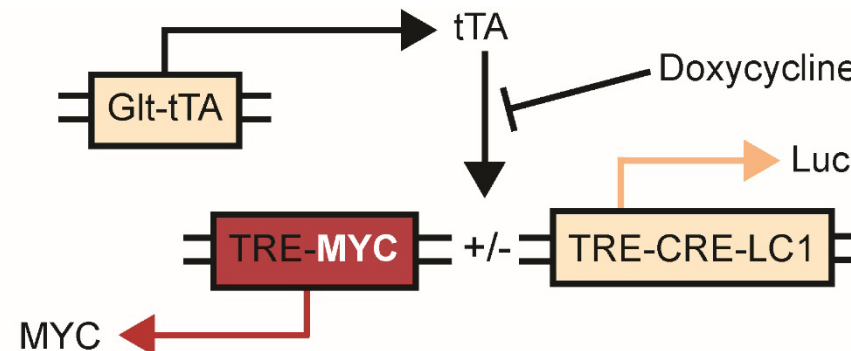


Swartling et al. Genes Dev. (2010)



*Northcott et al.
Nature. (2017)*

New **MYC**-driven transgenic model of Group 3 MB (**GMYC**)



*Mainwaring et al.
Nature Communications. (2023)*

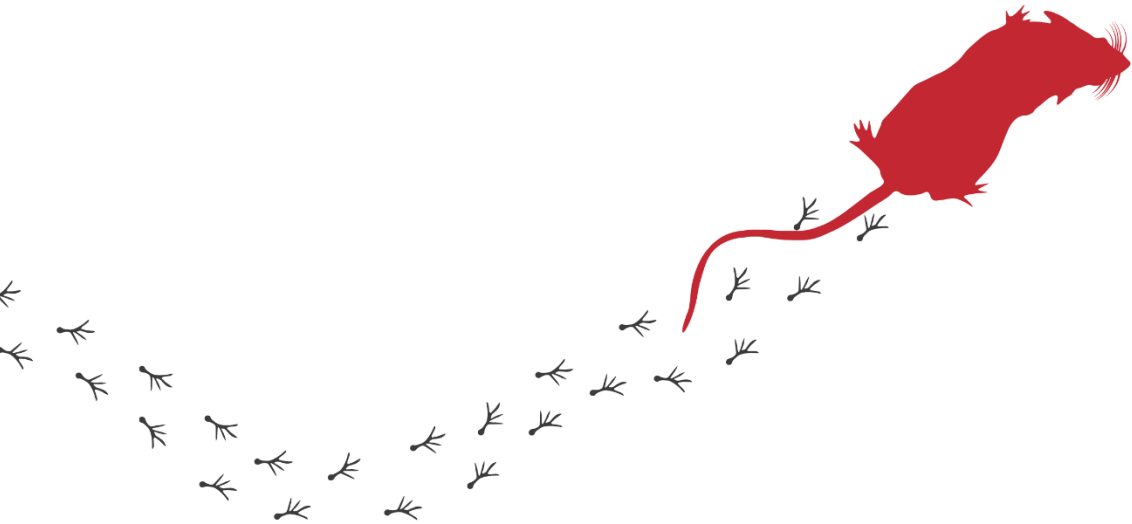


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Paper I

[*Manuscript and project title*
Paper I].

Author A., Author B., Author C., Author D., Author E., Author F., Author G., Author H., Author I., Author J., Author K., Author L., Author M., Author N., Author O., Author P., Author Q.





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[Paper I data and findings]





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Paper II

*[Manuscript and project title
Paper II]*

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Author K., Author L., Author M., Author N., Author O., Author P., Author Q.





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[Paper II data and findings]





Summary



- **Paper 1** – [*A short overview of the project findings and clinical relevance was provided here as a take-home message*]
- **Paper 2** – [*A short overview of the project findings and clinical relevance was provided here as a take-home message*]

